

RareCure AI — The Diagnostic & Treatment Intelligence Platform for Rare Diseases

Tagline: *Finally, answers for the undiagnosed.*

Executive Summary

400 million people worldwide suffer from rare diseases. The average patient waits 5-7 years for diagnosis, sees 8+ specialists, and receives 2-3 misdiagnoses before finding answers. This diagnostic odyssey costs the US healthcare system \$1 trillion annually in unnecessary tests and treatments.

RareCure AI is the autonomous intelligence platform that slashes rare disease diagnosis time from years to weeks. By combining multi-modal AI analysis, the world's largest rare disease knowledge graph, and direct connections to specialists and clinical trials, we transform the rare disease patient journey from nightmare to navigable.

The Problem

The Rare Disease Paradox

- **7,000+ identified rare diseases** — but individually rare means collectively massive
- **1 in 10 people** affected globally (400M patients)
- **95% have no FDA-approved treatment**
- **50% are children**, with 30% dying before age 5
- **Average diagnostic odyssey: 5-7 years**
- **\$1 trillion** annual US healthcare cost from diagnostic delays

Why Traditional Healthcare Fails

1. **Knowledge Fragmentation:** No single doctor can know 7,000+ diseases
2. **Symptom Overlap:** Rare diseases mimic common conditions
3. **Geographic Dispersion:** Specialists are scattered globally
4. **Data Silos:** Medical records don't follow patients
5. **Research Gaps:** 95% of rare diseases have no approved therapies
6. **Economic Neglect:** Small patient populations = limited pharma interest

The Human Cost

"We spent 6 years and \$300,000 going doctor to doctor. Our daughter was diagnosed with three different conditions before someone finally found the answer. By then, irreversible damage had been done." — Parent of child with undiagnosed genetic condition

The Solution

RareCure AI: Your Diagnostic Co-Pilot

For Patients & Families: - Upload medical records, genetic data, symptoms, family history - AI analyzes across 7,000+ rare diseases simultaneously - Receive prioritized diagnostic hypotheses with confidence scores - Connect directly with top specialists who've seen your condition - Get matched to relevant clinical trials worldwide - Join peer communities of others on the same journey

For Physicians: - Diagnostic decision support trained on rare disease literature - Real-time access to case-matched specialists for consultation - Automated literature monitoring for emerging research - Clinical trial matching for patients - Continuing education on rare conditions

For Researchers & Pharma: - Aggregated (de-identified) rare disease data for research - Patient cohort identification for clinical trials - Real-world evidence generation - Natural history study infrastructure - Regulatory-grade data collection

Product Architecture

1. Diagnostic Intelligence Engine

RARECURE DIAGNOSTIC AI

Medical Records	Genetic Data	Phenotype Analysis
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Multi-Modal Fusion Engine
(Clinical + Genetic + Imaging)

Rare Disease Knowledge Graph
(7,000+ diseases, 10M+ papers,
1M+ patient journeys)

Hypothesis Generation & Ranking
(Explainable AI with citations)

Specialist & Trial Matching

2. Knowledge Graph Components

- **Disease Ontology:** 7,000+ rare diseases with full phenotypic profiles
- **Genetic Database:** 5,000+ disease-gene associations
- **Symptom Mapping:** 25,000+ symptom-disease relationships
- **Literature Corpus:** 10M+ papers continuously ingested
- **Patient Journeys:** 1M+ de-identified diagnostic odysseys
- **Expert Network:** 50,000+ rare disease specialists globally
- **Trial Registry:** Real-time sync with 500+ trial databases worldwide

3. Platform Modules

Module	Function	Key Features
DiagnosticAI	Hypothesis generation	Multi-modal analysis, confidence scoring, explainable reasoning
GeneticLink	Genomic interpretation	WGS/WES analysis, variant classification, family cascade
SpecialistConnect	Expert matching	Video consults, case collaboration, second opinions
TrialFinder	Clinical trial matching	Real-time eligibility, site logistics, enrollment support
CommunityHub	Patient connection	Peer support, shared experiences, advocacy tools
ResearchPortal	Data collaboration	Cohort identification, natural history, RWE generation

Technology Deep Dive

Multi-Modal Diagnostic AI

Our diagnostic engine processes:

1. **Clinical Data**
 - Medical records (HL7 FHIR, C-CDA)
 - Lab results with trending
 - Imaging studies (radiology AI integration)
 - Specialist notes and assessments
2. **Genetic Data**
 - Whole genome/exome sequencing
 - Panel testing results
 - Variant of uncertain significance (VUS) reclassification
 - Polygenic risk scores for complex presentations
3. **Phenotypic Data**
 - Structured symptom capture (HPO terms)
 - Natural language processing of patient descriptions
 - Photo/video analysis for visible manifestations
 - Temporal symptom evolution tracking
4. **Environmental & Family Data**
 - Family history with pedigree analysis
 - Environmental exposure assessment
 - Consanguinity detection
 - Geographic disease clustering

Explainable AI for Medical Trust

Every diagnostic hypothesis includes: - **Confidence Score**: 0-100% with uncertainty quantification - **Evidence Chain**: Specific symptoms, labs, genetic findings supporting hypothesis - **Literature Citations**: Relevant case reports and research papers - **Similar Cases**: De-identified similar patient journeys - **Recommended Actions**: Next diagnostic steps with expected information gain - **Specialist Recommendations**: Experts who've published on or treated this condition

Privacy-First Architecture

PRIVACY ARCHITECTURE

Patient Data	Federated Learning	Research Access
Encrypted at rest & in transit	Model trains on distributed data, never centralized	Differential privacy + k-anonymity
• HIPAA compliant • SOC 2 Type II • GDPR ready	• Zero-knowledge proofs • Patient-controlled sharing • Audit trails	

Market Opportunity

Total Addressable Market (TAM)

Segment	Market Size	Notes
Rare Disease Diagnostics	\$8.5B	Growing 12% annually
Genetic Testing	\$22B	Declining costs driving adoption
Rare Disease Therapeutics	\$242B	12% of global pharma market
Clinical Trial Services	\$65B	Rare disease trials fastest growing
Total Addressable	\$337B	

Serviceable Addressable Market (SAM)

Segment	Market Size	Our Focus
Diagnostic Odyssey Reduction	\$15B	Direct patient + payer value
Clinical Trial Recruitment	\$8B	Pharma partnerships
Rare Disease Data/RWE	\$5B	Research licensing
Specialist Network Services	\$2B	Consultation fees
Serviceable Addressable	\$30B	

Serviceable Obtainable Market (SOM) — Year 5

Revenue Stream	Target	Revenue
Patient Subscriptions	500K families	\$300M
Health System Licenses	200 systems	\$150M
Pharma Partnerships	50 programs	\$200M
Research Data Licensing	100 institutions	\$100M
Specialist Network Fees	10K consultations/month	\$50M
Year 5 Revenue		\$800M

Business Model

Revenue Streams

1. Patient & Family Plans

Tier	Price	Features
Discovery	Free	Basic symptom analysis, community access
Diagnostic	\$99/mo	Full AI analysis, unlimited record uploads
Concierge	\$499/mo	Specialist matching, care coordination, advocacy support
Enterprise Family	Custom	Multi-generational genetic analysis, cascade testing

2. Health System Partnerships

- **Per-diagnosis licensing:** \$500-2,000 per case
- **Enterprise SaaS:** \$500K-2M annually
- **Outcomes-based contracts:** Shared savings from reduced diagnostic odyssey costs

3. Pharma & Biotech

- **Trial recruitment:** \$5,000-15,000 per enrolled patient
- **Real-world evidence:** \$2-10M per study
- **Natural history studies:** \$1-5M annually
- **Companion diagnostic development:** Revenue share

4. Research & Academic

- **Data access licensing:** \$100K-500K annually
- **Cohort identification services:** \$50-200 per patient match
- **Collaborative research partnerships:** Grant-funded + IP share

Unit Economics

Metric	Value
Patient CAC	\$150 (digital) / \$50 (referral)
Patient LTV	\$2,400 (24-month average journey)
LTV:CAC	16:1 (referral) / 16:1 (digital)
Gross Margin	78%
Net Revenue Retention	115% (family expansion)

Go-To-Market Strategy

Phase 1: Disease-Specific Beachheads (Months 1-12)

Target Conditions: 1. **Undiagnosed genetic syndromes** — Broadest need, clearest value prop 2. **POTS/Dysautonomia** — Growing awareness, underserved, strong communities 3. **Rare autoimmune conditions** — Complex overlapping symptoms 4. **Mitochondrial diseases** — Severe, pediatric-heavy, diagnostic challenge 5. **Connective tissue disorders** — EDS, Marfan, etc. — Large communities

Launch Strategy: - Partner with top 5 patient advocacy organizations per condition - Recruit 10 key opinion leader physicians per disease area - Content marketing: “The Diagnostic Journey” documentary series - Free tier drives viral growth within patient communities

Phase 2: Health System Integration (Months 12-24)

Target Partners: - Academic medical centers with rare disease programs - Children's hospitals (50% of rare diseases are pediatric) - Genetic testing laboratories - Undiagnosed disease programs (NIH UDP model)

Integration Approach: - EHR integration (Epic, Cerner) - Clinical decision support at point of care - Specialist referral network enrollment - Outcomes tracking and reporting

Phase 3: Pharma Ecosystem (Months 18-36)

Value Propositions: - **Patient finding:** Identify undiagnosed patients for trials - **Site selection:** Data-driven trial site optimization - **Endpoint refinement:** Natural history data for regulatory submissions - **Post-market:** Real-world evidence for label expansion

Target Partners: - Rare disease-focused biotechs (BioMarin, Alexion, Ultragenyx) - Big pharma rare disease divisions (Sanofi Genzyme, Takeda) - Gene therapy companies (Spark, Bluebird Bio successors)

Phase 4: Global Expansion (Months 30-48)

Priority Markets: 1. **UK/EU:** Centralized healthcare, strong rare disease policies 2. **Japan:** Advanced healthcare, aging population, rare disease focus 3. **Middle East:** Consanguinity = high rare disease prevalence 4. **India:** 70M+ rare disease patients, growing healthcare access

Competitive Landscape

Current Solutions & Limitations

Competitor	Approach	Limitation
Invitae/GeneDx	Genetic testing only	No clinical correlation, no journey support
FindZebra	Diagnostic search tool	Basic matching, no patient support
FDNA/Face2Gene	Facial analysis	Single modality, narrow use case
TrialSpark/IQVIA	Trial recruitment	No diagnostic assistance
Health systems	Undiagnosed programs	Geographic limits, capacity constrained

RareCure Advantages

1. **Multi-Modal Integration:** Genetics + clinical + phenotype + imaging
2. **End-to-End Platform:** Diagnosis through treatment, not just one step
3. **Patient-Centric:** Patients own their data and journey
4. **Global Scale:** Not limited to single health system
5. **Continuous Learning:** Every case improves the model
6. **Community Power:** Network effects from patient community

Competitive Moats

DEFENSIBILITY STACK

DATA NETWORK EFFECTS

Every patient journey improves diagnostic accuracy

1M+ journeys = 10x better than competitors

EXPERT NETWORK

50K+ rare disease specialists globally

Exclusive consult relationships

PATIENT COMMUNITY

Trust built over diagnostic journeys

Community-to-community referrals

REGULATORY/CLINICAL VALIDATION

FDA breakthrough device designation

Peer-reviewed diagnostic accuracy studies

Traction & Validation

Early Indicators (Pre-Launch)

- **LOIs from 15 patient advocacy organizations** representing 2M+ patients
- **Advisory board** includes former FDA rare disease division director
- **Research partnership** with top 3 academic rare disease programs
- **Pilot interest** from 5 rare disease-focused biotechs
- **Waitlist:** 25,000 families registered for early access

Key Milestones

Milestone	Timeline	Status
Knowledge graph v1 (1,000 diseases)	Q3 2026	In progress
First patient pilot (n=500)	Q4 2026	Recruiting
FDA breakthrough device pre-submission	Q1 2027	Planned
Health system pilot (3 sites)	Q2 2027	LOIs signed
First pharma partnership	Q3 2027	Discussions active
Public launch	Q4 2027	Planned

Team

Leadership

CEO — [TBD] - Former Chief Medical Officer at leading genetic testing company - MD/PhD, rare disease research background - Built and scaled \$500M+ diagnostic business

CTO — [TBD] - Former ML Director at DeepMind Health - PhD in biomedical AI - Led diagnostic AI development for rare disease detection

Chief Medical Officer — [TBD] - Director of Undiagnosed Disease Program at major academic center - 20+ years rare disease clinical experience - Published 200+ papers on diagnostic approaches

Chief Data Officer — [TBD] - Former Head of Data Science at Illumina - Expert in genomic data infrastructure - Built industry’s largest genomic knowledge base

Advisory Board

- Former FDA Rare Disease Division Director
- Founder of leading rare disease patient advocacy organization
- Partner at top healthcare-focused VC
- CEO of successful rare disease biotech (acquired for \$2B+)
- Professor of Medical Genetics at Harvard/Stanford/Johns Hopkins

Financial Projections

Revenue Growth

Year	Revenue	Key Drivers
2027	\$5M	Pilot programs, early patient adoption
2028	\$35M	Health system launches, first pharma deals
2029	\$120M	Scale patient platform, expand pharma partnerships
2030	\$350M	International expansion, enterprise growth
2031	\$800M	Market leadership, platform network effects

Key Metrics Trajectory

Metric	2027	2028	2029	2030	2031
Registered Patients	50K	200K	600K	1.5M	3M
Diagnoses Facilitated	2K	15K	50K	150K	400K
Specialist Network	5K	15K	30K	50K	75K
Health System Partners	10	50	150	300	500
Pharma Partnerships	5	20	50	100	150

Path to Profitability

- **Gross margin:** 78%+ (software + services)
- **Operating margin target:** 25%+ at scale
- **Cash flow positive:** Year 4 (2030)
- **Capital efficiency:** \$1 raised = \$4 revenue at maturity

Funding & Use of Proceeds

Current Raise: Series A — \$40M

Use	Allocation	Purpose
R&D / AI Development	45%	Knowledge graph, diagnostic engine, platform
Clinical Operations	25%	Medical team, regulatory, validation studies
Go-to-Market	20%	Patient acquisition, health system sales, advocacy partnerships
G&A	10%	Team, operations, infrastructure

Funding Roadmap

Round	Timing	Amount	Purpose
Seed	Completed	\$8M	MVP, initial disease coverage
Series A	Current	\$40M	Platform launch, clinical validation
Series B	2028	\$100M	Scale, international, enterprise
Series C	2030	\$200M	Market dominance, platform expansion

Target Investors

- **Healthcare-focused VCs:** a16z Bio, GV, Andreessen, ARCH
- **Crossover funds:** T. Rowe Price, Fidelity, Wellington
- **Strategic:** Pharma corporate VCs, diagnostic company CVCs
- **Impact:** Patient-focused foundations, mission-aligned capital

Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Regulatory uncertainty	Medium	High	Early FDA engagement, CE marking in parallel
Diagnostic accuracy challenges	Medium	High	Rigorous validation, explainable AI, human-in-loop
Data privacy concerns	Medium	Medium	Privacy-first architecture, patient data ownership
Health system adoption friction	Medium	Medium	EHR integration, outcomes-based pricing
Competition from big tech	Low	High	Domain expertise moat, clinical validation, trust
Reimbursement challenges	Medium	Medium	Value-based contracts, outcomes data

Impact Metrics

Our North Star: Ending the Diagnostic Odyssey

By 2031, we will:

- **Diagnose 400,000+ patients** who would otherwise remain undiagnosed
- **Reduce average diagnosis time** from 5 years to 6 months
- **Connect 1 million patients** to appropriate specialists and trials
- **Generate data** that leads to 50+ new rare disease treatments
- **Save \$10B+** in unnecessary healthcare spending

The Human Impact

Every diagnostic odyssey we shorten means: - A child who gets treatment before irreversible damage - A family that stops the exhausting search for answers - A patient who finds their community and support - A researcher who gets data to develop new therapies

This is healthcare's most broken journey. We're here to fix it.

Why Now?

Convergent Tailwinds

1. **AI Maturity:** Multi-modal foundation models can finally process diverse medical data
2. **Genetic Testing Costs:** WGS approaching \$100, democratizing access
3. **Data Availability:** EHR adoption enables unprecedented data aggregation
4. **Regulatory Support:** FDA rare disease focus, breakthrough designations
5. **Patient Activism:** Communities demanding better, connected via social media
6. **Pharma Investment:** Record rare disease R&D, need for patient finding
7. **Telehealth Normalization:** Remote specialist consults now accepted

The Window

The next 3 years will determine who builds the rare disease intelligence layer. Health systems are ready for AI. Patients are organized and demanding change. Pharma is investing billions. The infrastructure exists.

The question isn't whether this platform will be built. It's who will build it.

Call to Action

We're looking for:

1. **Investors** who understand healthcare transformation and patient impact
 2. **Physicians** passionate about diagnostic excellence for rare diseases
 3. **Patients & advocates** to co-design the platform you deserve
 4. **Pharma partners** committed to reaching rare disease populations
 5. **Technologists** building healthcare AI that actually works
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Contact: founders@rarecure.ai

Website: rarecure.ai

"In the time it took you to read this, three children were born with rare diseases that will take years to diagnose — if they're diagnosed at all. We're building the platform to change that."

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